Inverse Kinematics Solution for Six-DOF Serial Robots Based on BP Neural Network

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***Abstract*—The inverse kinematics solution of the Six-DOF Serial Robots is a highly complex nonlinear problem, and the existence of the solution is not unique, therefore, the inverse kinematics method has attracted extensive attention. In this paper, BP neural network is applied to the inverse kinematics of Six-DOF Serial Robots, and the process of inverse solution is transformed into training of network weights. The RBT-6T/S03S manipulator was used as the research object to establish the mathematical model, and the LM algorithm was used to train the network. Finally, the method is analyzed by MATLAB and the results show that the kinematic inverse solution can be obtained quickly and efficiently by using BP neural network.**

Keywords—Inverse Kinematics Solution; the Six-DOF Serial Robots; BP neural network

I. INTRODUCTION

The kinematics analysis of Six-DOF Serial Robots directly influences the study of Motion Planning and Trajectory Tracking Control. In contrast to the forward kinematic solution of Six-DOF Serial Robots, the inverse kinematics problem is a complex nonlinear problem, and the number of inverse solutions is not exclusive [1], but in the process of controlling the application, the robot often needs to select a set of optimal solutions when performing tasks. Therefore, the problem of inverse kinematics of multi-degree-of-freedom robot has been one of the focuses of robotics research.

The traditional methods of inverse solution mainly include analytic method, numerical iteration method, geometric method and so on. The inverse transform analysis proposed by Paul et al is very effective for the inverse solution of multi-degree-of-freedom robot, but the computation is too large and the choice of multiple solutions is also difficult to determine. Numerical iteration method depends on the initial point and can converge to a single solution, but it is difficult to guarantee the accuracy. Moreover, it is difficult to realize real-time control by the limitation of convergence speed. After that, some scholars put forward the method of Double Quaternions to solve the solution of inverse kinematics, which can effectively avoid the degradation problem in the inverse kinematics solution and reduce the type of the mechanism [2].

Artificial neural network is a network composed of many processing units interconnected, simulating human brain's information processing function, and it has strong nonlinear, self-organizing and self-learning ability [3]. In recent years, some scholars have used neural network algorithm to solve the problem of inverse solution of multi-degree-of-freedom robot, such as Shital S et al used RBF neural network to realize the inverse solution of Six-DOF Serial Robots [4]. In this paper, the BP neural network is applied to the inverse kinematics problem based on the kinematics of the Six-DOF Serial Robots. BP neural network is widely used in function approximation, data compression and so on. In addition, BP neural network has a strong nonlinear mapping ability, so in this paper, BP neural network is used to find the inverse solution of Six-DOF Serial Robots.

II. KINEMATICS ANALYSIS OF THE SIX-DOF SERIAL ROBOTS

This paper takes RBT-6T/S03S as the object of study, which is a kind of Six-DOF Serial Robots. In the analysis of Six-DOF Serial Robots, D-H method is often used to establish coordinate system, which strictly defines the coordinate system of each joint [5]. The following table shows the D-H parameter table for a Six-DOF Serial Robots:

TABLE1. D-H PARAMETERS OF RBT-6T/S03S ROBOT

Connecting Bar	**Angle of Rotation**	**Offset (mm)**	**Angle of Torsion**	**Bar length (mm**
1	(0)	(387.5)	(-90)	(0)
2	(-90)	(0)	(0)	(230)
3	(0)	(0)	(-90)	(107)
4	(0)	(268)	(-90)	(0)
5	(0)	(0)	(-90)	(0)
6	(0)	(165.5)	(0)	(0)

After analysis, according to the D-H parameter table, the Descartes coordinate system of the robot is established:

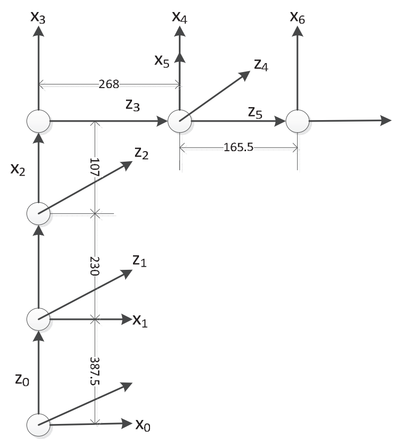

Fig.1 D-H coordinate of RBT-6T/S03S Manipulator

A. Kinematics Analysis of Six-DOF Serial Robots

The research of the positive kinematics of robot arm is that to solve the problem of the attitude of the manipulator end effector by giving the angles of each joint. According to the established D-H coordinates, the transformation matrix between the connecting rods can be obtained:

$$A_i = Rot(z,\theta_i)Trans(0,0,d_i)Trans(a_i,0,0)Rot(x,a_i) \quad (1)$$

$$T = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = A_1A_2A_3A_4A_5A_6 \quad (2)$$

Where

$$n_x = (c_1c_2c_3 - c_1s_2s_3)*(c_4c_5c_6 - s_4s_6) + s_1*(s_4c_5c_6 + c_4s_6) + (-c_1c_2s_3 - c_1s_2c_3)*s_5c_6$$

$$n_y = (s_1c_2c_3 - s_1s_2s_3)*(c_4c_5c_6 - s_4s_6) - c_1*(s_4c_5c_6 + c_4s_6) + (-s_1c_2s_3 - s_1s_2c_3)*s_5c_6$$

$$n_z = (-s_2c_3 - c_2s_3)*(c_4c_5c_6 - s_4s_6) + (s_2s_3 - c_2c_3)*s_5c_6$$

$$o_x = (c_1c_2c_3 - c_1s_2s_3)*(-c_4c_5s_6 - s_4c_6) + s_1*(-s_4c_5s_6 + c_4c_6) - (-c_1c_2s_3 - c_1s_2s_3)*s_5s_6$$

$$o_y = (s_1c_2c_3 - s_1s_2s_3)*(-c_4c_5s_6 - s_4c_6) - c_1*(-s_4c_5s_6 + c_4c_6) - (-s_1c_2s_3 - s_1s_2s_3)*s_5s_6$$

$$o_z = (-s_2c_3 - c_2s_3)*(-c_4c_5s_6 - s_4c_6) - (s_2s_3 - c_2c_3)*s_5s_6$$

$$a_x = -(c_1c_2c_3 - c_1s_2s_3)*c_4s_5 - s_1s_4s_5 + (-c_1c_2s_3 - c_1s_2c_3)*c_5$$

$$a_y = -(s_1c_2c_3 - s_1s_2s_3)*c_4s_5 + c_1s_4s_5 + (-s_1c_2s_3 - s_1s_2c_3)*c_5$$

$$a_z = -(-s_2c_3 - c_2s_3)*c_4s_5 + (s_2s_3 - c_2c_3)*c_5$$

$$p_x = -d_6*(c_1c_2c_3 - c_1s_2s_3)*c_4s_5 - d_6s_1s_4s_5 + (-c_1c_2s_3 - c_1s_2c_3)*(d_4 + d_6c_5) + a_3c_1c_2c_3 - a_3c_1s_2s_3 + a_2c_1c_2 + a_1c_1$$

$$p_y = -d_6*(s_1c_2c_3 - s_1s_2s_3)*c_4s_5 + d_6c_1s_4s_5 + (-s_1c_2s_3 - s_1s_2c_3)*(d_4 + d_6c_5) + a_3s_1c_2c_3 - a_3s_1s_2s_3 + a_2s_1c_2 + a_1s_1$$

$$p_z = -d_6*(-s_2c_3 - c_2s_3)*c_4s_5 + d_1 + (s_2s_3 - c_2c_3)*(d_4 + d_6c_5) - a_3s_2c_3 - a_3c_2s_3 - a_2s_2$$

In the equation above, C_i and S_i is short for $\cos\theta_i$ and $\sin\theta_i$, θ_i represents the normal angle between two connecting rods. d_i represents the relative position between two connecting rods.

B. Inverse Kinematics Analysis

The inverse kinematic problem is that the end of the manipulator in the reachable workspace range, finding the corresponding angle of each joint variable θ_i through given position and attitude matrix. According to the traditional analytic method that multiply A_1^{-1} at both ends of the equation:

$$A_1^{-1}T = A_2A_3A_4A_5A_6 \quad (3)$$

Then the solution is based on the equality of the corresponding elements of the matrix, and so on, each joint angle θ_i can be obtained.

$$\theta_1 = arctg^{-1}[(-p_y + d_6*a_y)/(-p_x + d_6*a_x)]$$

$$\theta_2 = arctg^{-1}\{[(p_{x1}{}^2 + p_{y1}{}^2)^{1/2}*(-d_4c_3 - a_3s_3) - p_{z1}*(-d_4s_3 + a_3c_3 + a_2)]/[(p_{x1}{}^2 + p_{y1}{}^2)^{1/2}*(-d_4s_3 + a_3c_3 + a_2) + p_{z1}*(-d_4c_3 - a_3s_3)]\}$$

$$\theta_3 = 2arctg^{-1}\{[-d_4 \pm (d_4^2 + a_3^2 - N^2)^{1/2}]/(N + a_3)\}$$

$$\theta_4 = arctg^{-1}\{[-(a_x*s_1 - a_y*c_1)]/[-(a_x*c_1c_{23} + a_y*s_1c_{23} - a_z*s_{23})]\}$$

$$\theta_5 = arctg^{-1}\{[a_z*s_{23}c_4 - a_x*(c_1c_{23}c_4 + s_1s_4) - a_y*(s_1c_{23}c_4 - c_1s_4)]/(-a_x*c_1s_{23} - a_y*s_1s_{23} - a_z*c_{23})\}$$

$$\theta_6 = arctg^{-1}\{[n_z*s_{23}s_4 - n_x*(c_1c_{23}s_4 - s_1c_4) - n_y*(s_1c_{23}s_4 + c_1c_4)]/[o_z*s_{23}s_4 - o_x*(c_1c_{23}s_4 - s_1c_4) - o_y*(s_1c_{23}s_4 + c_1c_4)]\}$$

In the equation above:

$$N = a_3c_3 - d_4s_3$$

$$p_{x1} = c_1*(-d_4*s_{23} + a_3c_{23} + a_2c_2)$$

$$p_{y1} = s_1*(-d_4s_{23} + a_3c_{23} + a_2c_2)$$

$$p_{z1} = -d_4c_{23} - a_3s_{23} - a_2s_2$$

$$S_{23} = S_2C_3 + S_3C_2$$

$$C_{23} = C_2C_3 - S_2S_3$$

According to the formula, we can determine, according to the existence uniqueness of the inverse kinematics analytic method. There are many factors in the selection of results in inverse kinematics, for example, a set of solutions can be selected according to the principle of the shortest path of the robot to reach the target point [6].

III. INVERSE KINEMATICS BASED ON BP NEUTAL NETWORK

From the kinematics analysis of Six-DOF Serial Robots, it can be seen that the inverse solution obtained by analytic method is very complicated and the inverse solution is not unique, Therefore, in this section, the BP neural network is used to solve the inverse kinematics of the Six-DOF Serial Robots .

A. The Theory of BP Neural Network

The neural network is composed of a large number of neural units and as neural units needs three conditions [7]. The first is to have a group of synapses or joins, followed an accumulator with a response integration function is required, and finally through an incentive function to limit the output. The figure is a neuron model with n inputs:

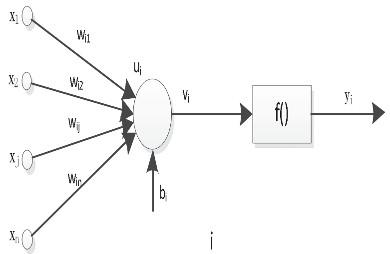

Fig 2. Artificial Neuron Model

In the figure, x_i is the input signal of the ith neuron unit. w_{ij} is called a weight. u_i is the output of the input signal after integration, b_i represents the threshold, v_i is a value that has been adjusted by bias, $f(\bullet)$ is the incentive function, y_i is the output signal of the ith neuron unit. Thus the output expression of the ith neuron is obtained:

$$y_i = f\left(\sum_{j=1}^{N} w_{ij} x_j + b_i\right) \qquad (4)$$

The general BP neural network is composed of input layer, hidden layer and output layer, including the forward propagation of the working signal and the back-propagation of the error signal [8]. The picture shows structure of BP neural network with single hidden layer. The first layer is the input layer, which is mainly composed of input signals; The second layer is a hidden layer, the number of hidden unit in this layer depends on the problem to be described; The third layer is the output layer, which is order to respond to the input signal.

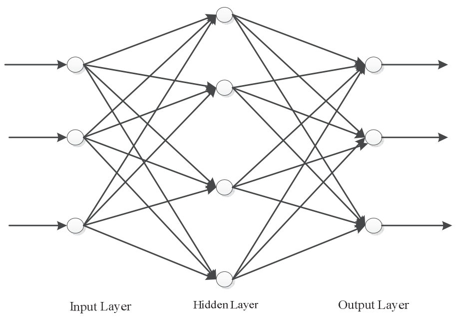

Fig 3. BP Neural Network Model

BP neural network learning is achieved by training samples, and the error between the output value and the target value is reduced through changing the weights of each connection, so as to achieve the purpose of building the model. The learning process consists of two main parts. First is the forward propagation of the work signal, the input signal from the input layer through the hidden layer to the output layer, and then produce the output signal, the network weight fixed in the process of forward communication. If the desired output cannot be obtained at the output layer, then it will turn to the stage of the reverse propagation of the error signal. The resulting error signal begins to propagate forward layer by layer from the output layer. In reverse propagation, the feedback of the error continuously affects the weight of the network and by modifying the weight of the network so that the actual output of the network and the expected output error is decreasing [9].

B. Training of Neural Networks

In the inverse kinematics calculation of the manipulator, the six joint angles $\theta_i = [\theta_1 \quad \theta_2 \quad \theta_3 \quad \theta_4 \quad \theta_5 \quad \theta_6]^T$ are obtained based on the position and orientation matrix T of the end effector. From the description of the matrix T, it can be concluded that the fourth row of T is always constant, so the first three lines in the matrix T can be used as the input of the BP neural network, and the joint angle θ_i is used as the output of the BP neural network. In this paper, the measurement is used to obtain the sample. During the Six-DOF Serial Robots reset operation, 300 points were recorded at each time and execute seven-times, finally 2100 sets of sample data were obtained. When the network is initialized, the target error is set at 10^{-6}, and the maximum number of iterations is 5000 times.

For the activation function of hidden layer, the hyperbolic tangent S type function is used in this paper and it can be expressed as:

$$f(v) = \frac{1-\exp(-x)}{1+\exp(-x)} \qquad (5)$$

The training algorithm adopts the gradient back propagation algorithm of Levenberg-Marquardt algorithm. LM algorithm is a combination of Newton method and gradient descent method. Compared with other algorithms, it has the advantages of fewer iterations, higher accuracy, shorter learning time and so on [10].

In the LM algorithm, the error index function is set as:

$$E(w) = \frac{1}{2}\sum_{i=1}^{n} \left\| y_i - y_i^{'} \right\|^2 = \frac{1}{2}\sum_{i=1}^{n} e_i^{\,2}(w) \qquad (6)$$

In the equation above, y_i is the desired network output. y_i' is the actual network output, N is the number of inputs, W is the weight, $e_i(w)$ is the error. Suppose that the w^k represents the weight of the K iteration, the new weight is $w^{k+1} = w^k + \Delta w$. In the LM algorithm, the weights are calculated as follows:

$$\Delta w = \left[J^T(w)J(w) + \mu I\right]^{-1} J^T(w)e(w) \qquad (7)$$

In the equation above, I is the unit matrix, μ is the learning rate, and $J(w)$ is the Jacobian matrix:

$$J(w) = \begin{bmatrix} \frac{\partial e_1(w)}{\partial w_1} & \frac{\partial e_1(w)}{\partial w_2} & \cdots & \frac{\partial e_1(w)}{\partial w_n} \\ \frac{\partial e_2(w)}{\partial w_1} & \frac{\partial e_2(w)}{\partial w_2} & \cdots & \frac{\partial e_2(w)}{\partial w_n} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial e_n(w)}{\partial w_1} & \frac{\partial e_n(w)}{\partial w_2} & \cdots & \frac{\partial e_n(w)}{\partial w_n} \end{bmatrix} \qquad (8)$$

IV. SIMULATION RESULT

As is show in the figures below, figure 4 is the training error plot of BP neural network, figure 5 shows the relative error plot between the test output angle and the expected output angle, figure 6 shows the contrast between the test output and the expected output for six joint angles, figure 7 is the output MSE plot.

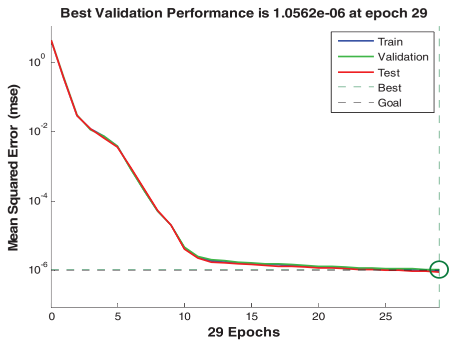

Fig 4. Training Error Plot of BP Neural Network

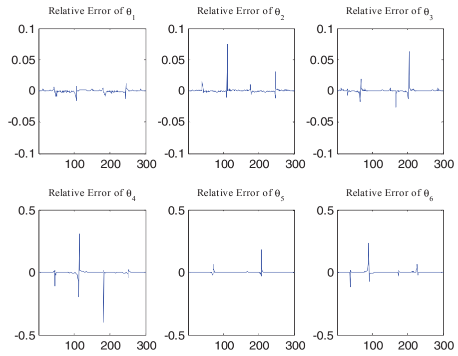

Fig 5. Relative Error Plot

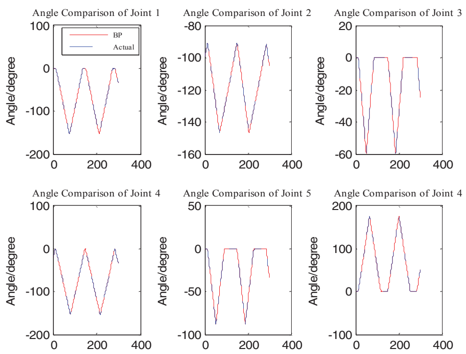

Fig 6. Contrast Between the Test Output and the Expected Output

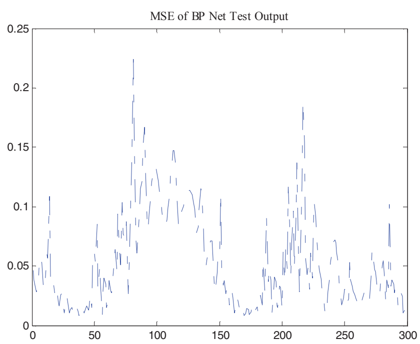

Fig 7. MSE Plot of BP Neural Network Test Output

It can be concluded from figure that 4.1 the neural network achieves the training error requirement at the twenty-ninth iteration, as can be seen from Figure 4.2 and Figure 4.3, the relative error of each joint angle is within 0.5 degrees, and there is no obvious error. Figure 4.4 shows that the MSE is below 0.25 degrees, indicating that trained BP neural network with high accuracy. In summary, the trained neural network is feasible in solving the inverse problem of six DOF manipulator.

V. CONCLUSION

In this paper, RBT-6T / S03S is used as the research object to establish the kinematics model of Six-DOF Serial Robots, and then design a BP neural network for the inverse kinematics method based on the LM algorithm for training the network, changing the network weights to achieve the inverse solution, so as to avoiding the complex operation. Through the simulation of the fourth part of the text, it can be seen that the method of inverse kinematics based on BP neural network is faster and can improve the real-time performance of control, and the accuracy of the solution is less than 0.5 degree, so that it can be quickly and effectively solved.

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